THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108: PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

Dec 21, 2018 9:30 am – 12:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. Under the Lantau Tomorrow Vision, a series of reclamation projects will be rolled out for the years to come.

   a) Explain the mechanism of two-stage selective tendering and discuss the pros and cons of this tendering approach over conventional selective tendering method. [15 marks]

   b) Describe the principle of serial tendering and discuss its suitability for developing a series of reclamation projects under the Lantau Tomorrow Vision. [10 marks]

2. The following table shows the duration and logical relationship of the activities involved in a construction project:

Activity ID	*Predecessor*	*Duration (days)*
A	--	10
B	A	30
C	B	30
D	C (SS10)	40
E	D (SS10)	40
F	C	40
G	D	28
H	E	30
I	D	30
J	E	30
K	G, H, I, J	70
L	D, E, F	60
M	K	100
N	L, M	10

   a) Prepare an activity-on-node network diagram based on the data given in the table above and identify the critical path of this project. [10 marks]

   b) Prepare a table showing the early and late start and completion dates of each activity and work out the total and free floats accordingly. [8 marks]

   c) Describe the measures to be taken by project manager to monitor and control project progress. [7 marks]

3. a) An overseas company has just been awarded a mega infrastructure project in Hong Kong. While this is their first and only project in Hong Kong so far, the company is not quite sure which organisational setting should be adopted. Highlight the features of functional and project-based organisational settings and explain their pros and cons. [16 marks]

   b) While matrix organisation is commonly employed by construction companies in general, examine whether this approach is a more appropriate option for this company under the current scenario. [9 marks]

4. a) An investor is planning to build a 200-unit service apartment. The usable floor area of this service apartment will be 9,000m^2 while the circulation area will take up 20% of the total floor space. The construction cost is estimated to be \$10,000/m^2 and the building will take 2 years to construct. At a monthly rent of \$25,000/unit, the service apartment will attract 90% occupancy during its first three years of operation before attaining a full occupancy rate from the fourth year onward. The initial yield for this type of development would be 6%. With the cost of advertisement being \$1,000,000 and a short-term finance of 12% per annum, calculate the maximum site value based on the above parameters and your engineering judgement. [19 marks]

   b) Distinguish the differences between tender price index and building cost index and give an example on their application in the construction industry. [6 marks]

5. a) Describe the roles and responsibilities of a civil engineer throughout the building life cycle. [17 marks]

   b) Discuss with examples the potential problems when tenderers are selected from the approved list of contractors. What action(s) should be taken when these problems are discovered? [8 marks]

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